

Endoscopic Utilization and Outcomes of Upper Gastrointestinal Bleeding in COVID-19: A Global Real-World Database Analysis

Running title: Endoscopy utilization for upper GI bleed in COVID-19

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Endoscopic Utilization and Outcomes of Upper Gastrointestinal Bleeding in COVID-19: A Global Real-World Database Analysis

Short title: Endoscopy utilization for upper GI bleed in COVID-19

Abstract

Introduction- Upper gastrointestinal bleeding (UGIB) events are increasingly recognized in the Coronavirus Disease-2019 (COVID-19) patients. The data about the rates of utilization of endoscopy and outcomes among these patients are scarce.

Methods- All hospitalized COVID-19 patients with UGIB (hematemesis or melena or both) were studied from January 1, 2020, to September 1, 2020, from the TriNetX multicenter research network database. Patients were divided into two groups- patients who underwent any endoscopies (endoscopy group) and did not undergo endoscopy (non-endoscopy group). A subgroup analysis was performed for patients with variceal (V-UGIB) and non-variceal UGIB (NV-UGIB). Mortality and development of acute respiratory distress syndrome were the primary outcomes of the study. Secondary outcomes were cross-section imaging (CT scan of the abdomen) and mechanical ventilation.

Results- A total of 3466 hospitalized COVID-19 had UGIB events from January 1, 2020, to September 1, 2020. 3.1% (3466/112,158). Only 51.1% of patients (1773/3446) underwent an endoscopy. The endoscopy group had a lower risk of mortality (risk ratio [RR] 0.63, 95%CI:0.52–0.77), and ARDS (RR 0.39, 95%CI:0.22–0.71). Rates of mechanical ventilation (RR 0.66, 95%CI:0.47–0.91) were lower, and the rate of CT scan utilization (RR 1.20, 95%CI:1.01–1.42) were higher in the endoscopy group. No differences in the outcomes were noted in V-UGIB and NV-UGIB groups.

Conclusion- UGIB events in hospitalized adult COVID-19 patients was 3.1%. Endoscopy utilization in this cohort was associated with a significantly lower risk of mortality (37%) and ARDS (61%). Future prospective studies are needed to validate these findings.

Keywords: Gastrointestinal bleeding; variceal bleeding; bleeding; upper GI bleeding; COVID-19; SARS-CoV-2; endoscopy; ARDS; mechanical ventilation; outcomes; TriNetX.

Endoscopic Utilization and Outcomes for Upper gastrointestinal GI Bleeding in COVID-19: A Global Real-World Database Analysis

Introduction

Upper GI bleeding (UGIB) is one of the most common GI emergencies needing hospitalization (1). It is associated with high mortality and morbidity if appropriate and timely interventions are not performed. Due to the Coronavirus Disease-2019 (COVID-19) pandemic, the number of UGIB admissions has reduced up to 40.7% (predominately non-variceal bleeding) without a change in the rates of variceal bleeding events (2). Though the reason for these findings remains obscure, it could be due to the stay-at-home orders issued by the public health officials and fear of contracting the disease at healthcare facilities (3). Furthermore, multiple national and international GI and endoscopic societies recommended deferring elective and semi-elective procedures, which led to a decline in the rates of endoscopic utilization throughout the globe.

COVID-19, which is caused by novel severe acute respiratory syndrome coronavirus-2 (SARS-CoV-2), can lead to pulmonary and extrapulmonary manifestations (4). GI manifestations, such as anorexia, nausea, vomiting, abdominal pain, taste changes, and diarrhea, are common in these patients. Atypical presentations of COVID-19 such as GI bleeding (GIB) have been reported as well (3, 5, 6). Most of the studies evaluating GIB are restricted to few case series and small retrospective studies (6). The data about the rate of UGIB in patients with COVID-19 is scarce. Furthermore, endoscopy utilization in these patients and related outcomes needs further evaluation since UGIB events in hospitalized patients can lead to significant morbidity and mortality.

Additionally, no study has evaluated the rate of UGIB in hospitalized COVID-19 patients. Hence, we aim to evaluate the rate of UGIB and outcomes in hospitalized COVID-19 patients who underwent endoscopies from a multicentric global real-world practice database. Additionally, outcomes were also evaluated between the variceal (V-UGIB) and non-variceal (NV-UGIB) events.

Methods

Description of the Data Source

We utilized a federated cloud-based network (TriNetX), which provides access to anonymized data of electronic medical records (EMRs) from 50 health care organizations (HCOs) across the globe. The TriNetX network was queried on September 14, 2020, and deidentified dataset of patients with COVID-19 aged from 18 years to 90 years were identified between January 1, 2020, and September 1, 2020. Data on this network is obtained from academic medical centers, specialty physician services, and community hospitals. TriNetX has received a waiver from IRB since it does not contain any protected health information (PHI) or the HCO Information, and only includes aggregated counts and statistical summaries of de-identified information.

Selection of the patients

Using the TriNetX analytics platform, COVID-19 patients were identified based on the Centers for Disease Control and Prevention (CDC) coding guidelines. Patients were included if they had one or more of the following International Classification of Diseases, Ninth and Tenth Revisions, Clinical Modifications (ICD-10-CM) codes in the EMR's: U07.1 COVID-19; B34.2 Coronavirus infection, unspecified; B34.2 Coronavirus infection, unspecified and 9088 SARS coronavirus 2 and related RNA (1). All patients who received inpatient services from the above codes were included. The patients who developed K92.0 (hematemesis) or K92.1 (melena) or both one day before and within 30 days of hospital admission were included. These hospitalized COVID-19 patients who developed bleeding were divided into two groups- upper gastrointestinal bleeding (UGIB) with endoscopy (CPT 1007241 endoscopy procedures on the esophagus, CPT 1007635 flexible sigmoidoscopy, CPT 1022231 flexible colonoscopy) and without endoscopy.

Comorbidities were identified using codes I10 (essential hypertension), I20-25 (Ischemic heart disease), N18 (CKD), E08-13 (Diabetes Mellitus), J44.9 (Chronic obstructive pulmonary disease, unspecified), E65-E68 (Overweight, obesity and hyperalimentation) and K74 (fibrosis and cirrhosis of liver).

Study Outcomes

Outcomes of the study include death, acute respiratory distress syndrome (ARDS), utilization of CT scan of the abdomen & pelvis, and mechanical ventilation. The outcomes were measured before and after 1:1 propensity matching of the groups based on the baseline demographics and comorbidities.

Statistical Analysis

All statistical analyses were performed on the TriNetX online platform. To safeguard the PHI, TriNetX conceals patient counts by rounding patient counts in analyses up to the nearest 10. Descriptive measures such as means with standard deviations and proportions were used to describe patient characteristics. For each outcome, the risk ratio was calculated to estimate the risk of UGIB events on the outcomes. A priori defined 2-sided alpha value of $<.05$ was used for statistical significance.

Propensity Score Matching: To account for confounding variables, we performed a 1:1 propensity score matching with a caliper of 0.1 pooled standard deviations based on the following variables: age, gender, race, hypertension, diabetes mellitus, obesity, chronic kidney disease (CKD), Ischemic heart diseases, chronic obstructive pulmonary disease and cirrhosis of the liver.

Results

A total of 3466 adult hospitalized COVID-19 patients had UGIB events from January 1, 2020, to September 1, 2020. The rate of UGIB among hospitalized COVID-19 was 3.1% (3466/112,158). The rate of V-UGIB among UGIB was 3.9% (136/3466). A total of 1773 patients (51.1%) underwent endoscopy (endoscopy group) and 1693 (48.9%) did not undergo endoscopy (non-endoscopy group). Clinical manifestations at presentation, laboratory findings, and clinical outcomes of these patients are noted in Table 1.

The results of our analysis showed that the endoscopy group had a lower risk of mortality (risk ratio [RR] 0.63, 95%CI:0.52–0.77), and ARDS (RR 0.39, 95%CI:0.22–0.71). Secondary outcomes such as mechanical ventilation (RR 0.66, 95%CI:0.47–0.91) were lower, and the rate of CT scan

utilization (RR 1.20, 95%CI:1.01–1.42) were higher in the endoscopy group. No differences in outcomes were noted in V-UGIB and NV-UGIB groups (table 2).

Discussion

This study showed that the rate of UGIB among hospitalized adult COVID-19 patients was low at 3.9%. Moreover, only about half (51.1%) of patients underwent endoscopies for evaluation for UGIB. As expected, we found that patients who underwent endoscopy had a 37% lower risk of mortality than those who did not, even after adjusting for potential confounders. Our results indicate the significance of endoscopy in the care of UGIB, even in COVID-19 infected patients.

Multiple studies have shown that COVID-19 patients are at risk of developing GIB during hospitalization (3, 5, 6). The exact mechanism of SARS-CoV-2 induced GI injury is unknown, but the SARS-CoV-2 virus may gain entry into the GI tract via angiotensin-converting enzyme (ACE-2) receptors present in the GI tract. Gastric acid is postulated to disable the virus due to the adverse effects of the acidic environment. Despite this, SARS-CoV-2 can enter the small intestine and exhibit its cytopathic effect on the enterocytes (8). Furthermore, acute hemorrhagic colitis has been reported in COVID-19(10).

Published data on GIB in COVID-19 patients indicate that endoscopy utilization is low, most likely due to the fear of viral transmission, conservation of personal protective equipment (PPE), and local hospital policies (3, 5, 6, 11). Nevertheless, UGIB could carry high morbidity and mortality if appropriate and timely measures are not undertaken (1). Our study showed that endoscopy utilization was 51.1%, with a 37% lower risk of mortality in these patients. Furthermore, rates of mechanical ventilation reduced by 34%, probably due to early endoscopy. Regardless, our findings are similar to other studies performed on non-COVID-19 patients showing that endoscopies decrease mortality in UGIB. Nevertheless, the decision to perform endoscopy in these patients is based on the clinical judgment and assessment of the risk and benefits of performing the procedure.

There are some limitations to our study. Potentially unrecognized confounding variables could affect the findings of our study. TriNetX obtains data from multiple HCOs stored in the form of

ICD-10 codes, which could lead to “coding errors.” TriNetX does not have information about omitted data entries. All variables were checked independently by two authors (AP and MG) to minimize these biases. We could not evaluate the timing of endoscopy and follow up of these patients after the discharge. Nevertheless, the availability of extensive sociodemographic history, clinical, laboratory, utilization endoscopy, mechanical ventilation, and mortality rates provide useful insights into the UGIB-related events in COVID-19 patients.

Conclusion

This study shows that the rate of UGIB events in hospitalized adult COVID-19 patients can be up to 3.9%. Endoscopy utilization was 51.1% in these patients. Patients who underwent endoscopy had a significantly lower risk of mortality (37%) and ARDS (61%). Larger prospective studies are needed to validate these findings.

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Table 1: Baseline characteristics and clinical outcomes in COVID-19 patients with upper gastrointestinal bleeding (UGIB) who underwent endoscopy (UGIB w Endo) compared to individuals without endoscopy (UGIB w/o Endo)

	Before Matching			After Matching [€]		
Characteristic Name	UGIB w Endo N= 1773	UGIB w/o Endo N= 1693	P-Value	UGIB w Endo N= 1602	UGIB w/o Endo N= 1602	P-Value
Age (SD)	62.19 (16.93)	62.27 (18.03)	0.8959	62.41 (16.93)	62.56 (17.91)	0.8055
Female	783 (44.16)	740 (43.71)	0.7882	699 (43.63)	701 (43.76)	0.9432
White	1155 (65.14)	1029 (60.78)	0.0078	1020 (63.67)	1020 (63.67)	1.0000
African American	352 (19.85)	427 (25.22)	0.0002	345 (21.54)	351 (21.91)	0.7971
Comorbidities						
Essential hypertension	1181 (66.61)	1091 (64.44)	0.1793	1027 (64.11)	1043 (65.11)	0.5544
Ischemic heart diseases	718 (40.50)	670 (39.58)	0.5799	628 (39.20)	644 (40.20)	0.5635
Diabetes mellitus	622 (35.08)	608 (35.91)	0.6094	568 (35.46)	578 (36.08)	0.7124
CKD	580 (32.71)	546 (32.25)	0.7714	513 (32.02)	519 (32.40)	0.8205
Obesity	541 (30.51)	468 (27.64)	0.0630	446 (27.84)	454 (28.34)	0.7532
COPD	338 (19.06)	317 (18.72)	0.7985	304 (18.98)	305 (19.04)	0.9641
Cirrhosis of liver	348 (19.63)	167 (9.86)	<0.0001	300 (18.73)	158 (9.86)	<0.0001
Clinical						
Gastritis and duodenitis	561 (31.64)	276 (16.30)	<0.0001	499 (31.15)	251 (15.67)	<0.0001
Gastric ulcer	382 (21.55)	111 (6.56)	<0.0001	338 (21.10)	104 (6.49)	<0.0001
Duodenal ulcer	247 (13.93)	62 (3.66)	<0.0001	218 (13.61)	57 (3.56)	<0.0001
Lab						
Hemoglobin (SD)	8.74 (2.54)	10.36 (2.80)	0.0000	8.73 (2.53)	10.37 (2.82)	<0.0001
Platelets (SD)	212.99 (129.06)	225.03 (121.75)	0.0068	214.70 (128.99)	224.77 (121.31)	0.0292

Blood Urea nitrogen (SD)	27.54 (24.15)	26.24 (22.80)	0.1203	27.58 (23.96)	26.37 (22.97)	0.1644
Creatinine (SD)	1.52 (2.48)	1.68 (1.98)	0.0542	1.53 (2.57)	1.66 (1.98)	0.1295
Prothrombin time (SD)	15.53 (6.60)	15.67 (6.47)	0.5535	15.39 (5.87)	15.68 (6.42)	0.2242
INR (SD)	1.34 (0.56)	1.36 (0.64)	0.1912	1.32 (0.50)	1.37 (0.64)	0.0448
Hemoglobin A1c (SD)	6.16 (1.69)	6.34 (1.71)	0.0159	6.20 (1.70)	6.35 (1.69)	0.0604
Neutrophils (SD)	207.66 (1116.04)	258.86 (1142.10)	0.2613	206.27 (1133.01)	265.13 (1162.87)	0.2222
Lymphocytes (SD)	16.13 (10.63)	17.00 (12.51)	0.0421	16.30 (10.75)	16.77 (12.25)	0.2862
Outcome			Risk Ratio			Risk Ratio
Death	159 (8.97)	235 (13.88)	0.65 (0.53 - 0.78)	141 (8.80)	224 (13.98)	0.63 (0.52 - 0.77)
ARDS	17 (0.96)	39 (2.30)	0.42 (0.24 - 0.73)	15 (0.94)	38 (2.37)	0.39 (0.22 - 0.71)
CT scan	285 (16.07)	219 (12.94)	1.24 (1.06 - 1.46)	248 (15.48)	207 (12.92)	1.20 (1.01 - 1.42)
MV	61 (3.44)	92 (5.43)	0.63 (0.46 - 0.87)	57 (3.56)	87 (5.43)	0.66 (0.47 - 0.91)

ARDS - Acute Respiratory Distress Syndrome; COPD – Chronic Obstructive Pulmonary Disease; MV – Mechanical Ventilation; CT – computed tomography; INR - international normalized ratio

€ A 1:1 propensity score matching was done based on the following variables: age, gender, race, hypertension, diabetes mellitus, obesity, chronic kidney disease (CKD), Ischemic heart diseases, chronic obstructive pulmonary disease, and cirrhosis of the liver.

Table 2: Clinical Outcomes in the subgroup analysis based on non-variceal bleeding (NVB) with Endo and non-variceal bleeding (NVB) without Endo after propensity matching.

	Before Matching	Before Matching	Risk Ratio (95% CI)	After Matching [€]	After Matching [€]	Risk Ratio (95% CI)
Primary Outcome	NVB w Endo N= 1435	NVB w/o Endo N= 1460		NVB w Endo N= 1322	NVB w/o Endo N= 1322	
Death	112 (7.80)	202 (13.84)	0.56 (0.45 - 0.70)	99 (7.49)	184 (13.92)	0.54 (0.43 - 0.68)
ARDS	14 (0.98)	29 (1.99)	0.49 (0.26 - 0.93)	13 (0.98)	24 (1.82)	0.54 (0.28 - 1.06)
Secondary outcomes						
CT scan	210 (14.63)	171 (11.71)	1.25 (1.04 - 1.51)	192 (14.52)	158 (11.95)	1.22 (1.00 - 1.48)
MV	48 (3.34)	86 (5.89)	0.57 (0.40 - 0.80)	44 (3.33)	78 (5.90)	0.56 (0.39 - 0.81)

Footnote: ARDS - Acute Respiratory Distress Syndrome; CI: Confidence interval; MV – Mechanical Ventilation; CT – computed tomography; NVB; Non-Variceal bleeding.